Lab 3: Simulation in COMSOL and Turbulence Modeling

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Deliverable 1: Reflection on Geometry

The geometry is a reasonable simplification to model a single breakout cooling hole since it captures the fact that the hole is small relative to the bulk fluid flow along the blade and that this fluid flows tangentially to the plane of the outlet hole. However, it overlooks the fact that the actual blade is curved and rotating as part of a turbine. Thus, there are likely centrifugal forces affecting flow that may not be taken into account. Additionally the curve of the blade could affect the shape of the velocity profile and boundary layer. Additionally, there are no properties established about the material the blades are made out of, only of the fluid moving along them. The surface roughness of the blades affects friction exerted on the fluid and, thus, the boundary layer. Additionally, the various outlet holes likely interact to affect the flow on the blade overall, which is not accounted for using a single hole. Finally, water is used instead of air. Despite using matching Reynold's numbers, the cooling effects would be different using water than air due to differing thermophysical properties (if thermal effects are being considered).

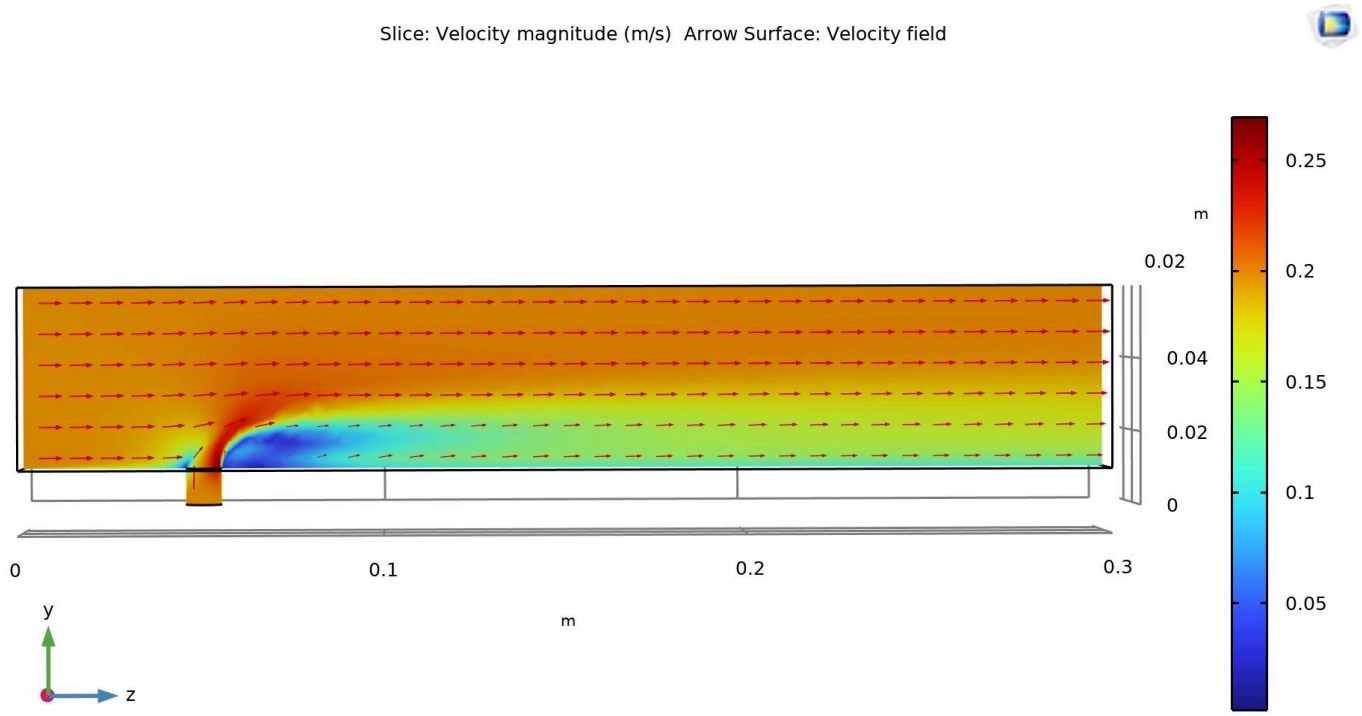
Accuracy of the model could be improved by adding a slight curve to the prism to account for the fact that the fluid flows along a curved blade and adding force to represent the blade's rotation. Additionally, specifying a blade material could give a better idea of the effects of the surface roughness on the boundary layer properties. Finally, though this would cause great increases in computing time, modelling multiple holes may be more accurate to show how flow at one hole is affected by another.

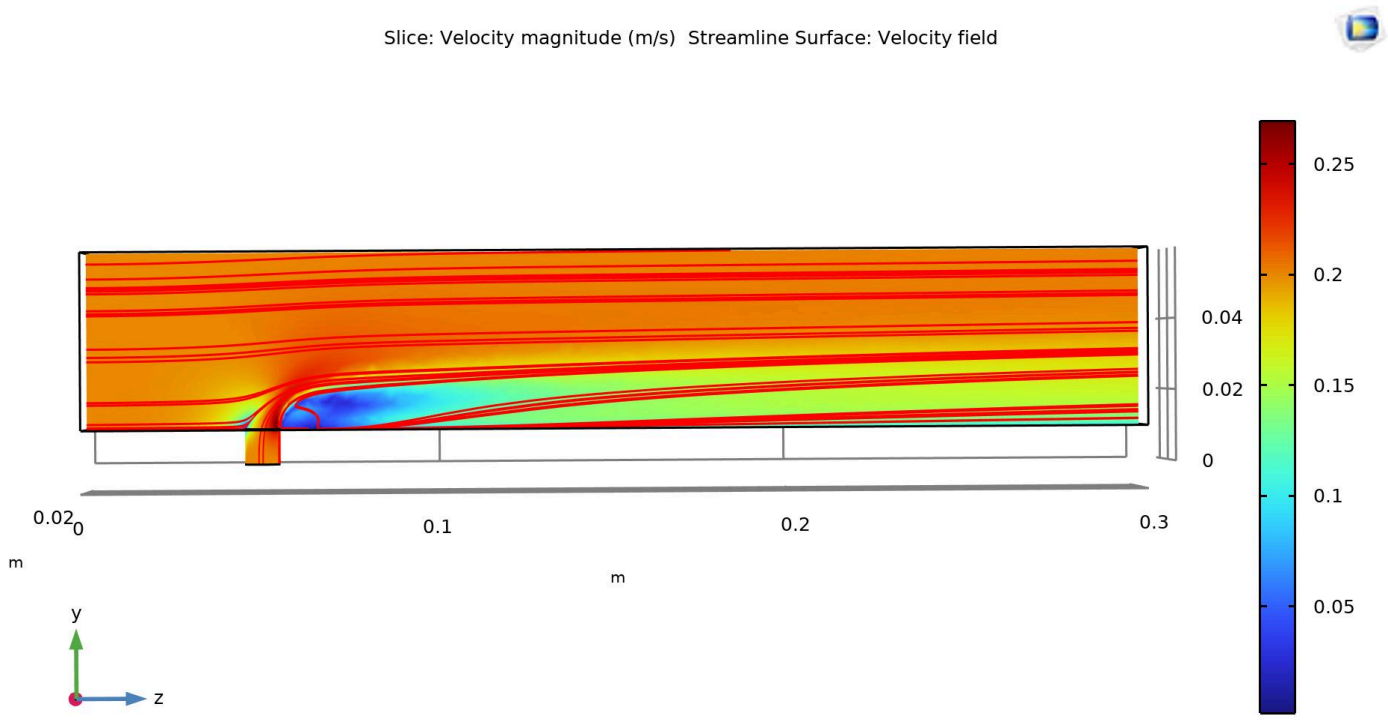
Deliverable 2: Computation time for Solution

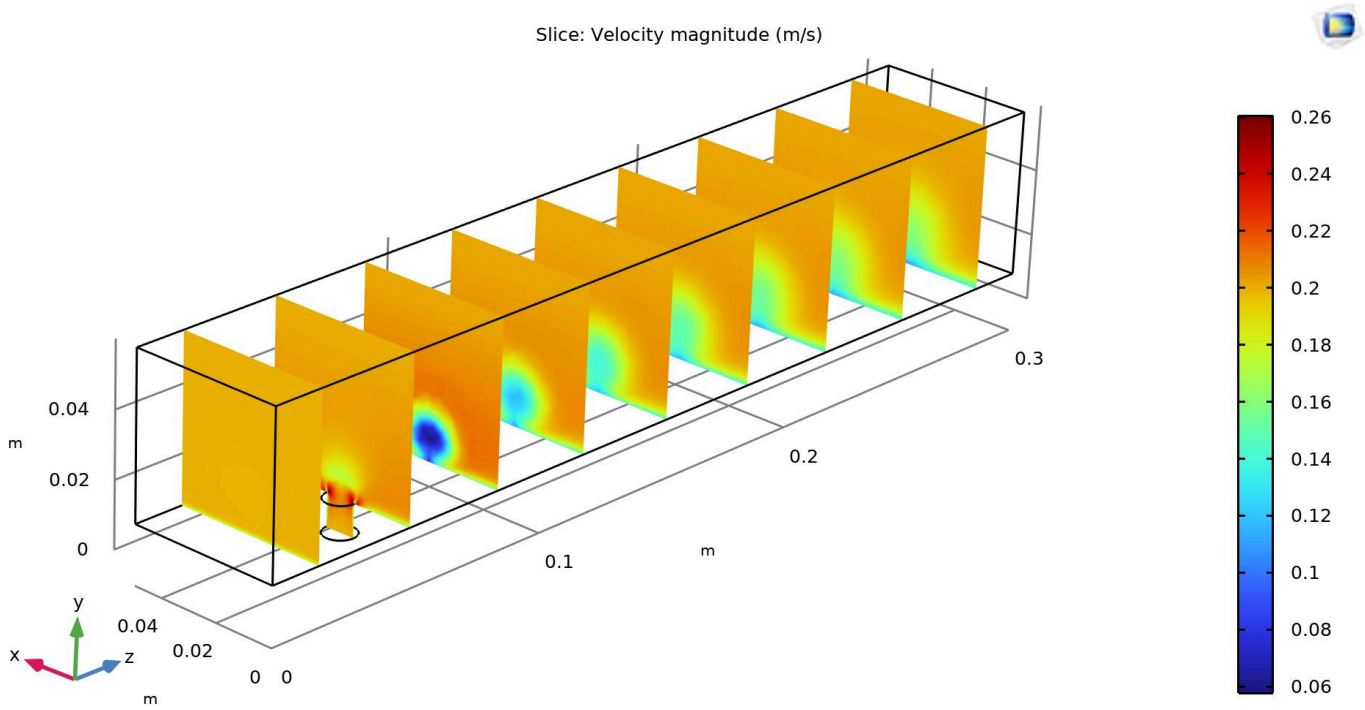
k-ε Turbulence Model: 3 minutes, 31 seconds (211 seconds)
k-ω Turbulence Model: 3 minutes, 9 seconds (189 seconds)
Algebraic y-Plus Turbulence Model: 1 minute, 25 seconds (85 seconds)

Deliverable 3: Data Analysis

Velocity Plots:







Brief Description of Flow Field:

Fluid flow is relatively uniform and horizontal before the breakout cooling hole. Velocity immediately downstream from the breakout cooling hole is significantly reduced, around 0.007 m/s, near the opening (bottom of the blade) and relatively unchanged further away vertically from the hole. In other words, there is a separated region directly downstream of the hole where mixing of the introduced fluid occurs. There is another separated region just before the breakout hole/entering fluid since the bulk flow is redirected away from that corner, here the velocity is around 0.0017 m/s. Moving further downstream, the velocity gradually regains uniformity. Directly above the separated region, there is a region where the flow speeds are very fast, around 0.2695 m/s, which I think about as where the bulk fluid is compressed to a smaller flow area due to the separated region. This hotspot occurs at the downstream edge of the breakout cooling hole. Throughout the volume, flow is mostly horizontal but has a significant vertical component near the breakout cooling

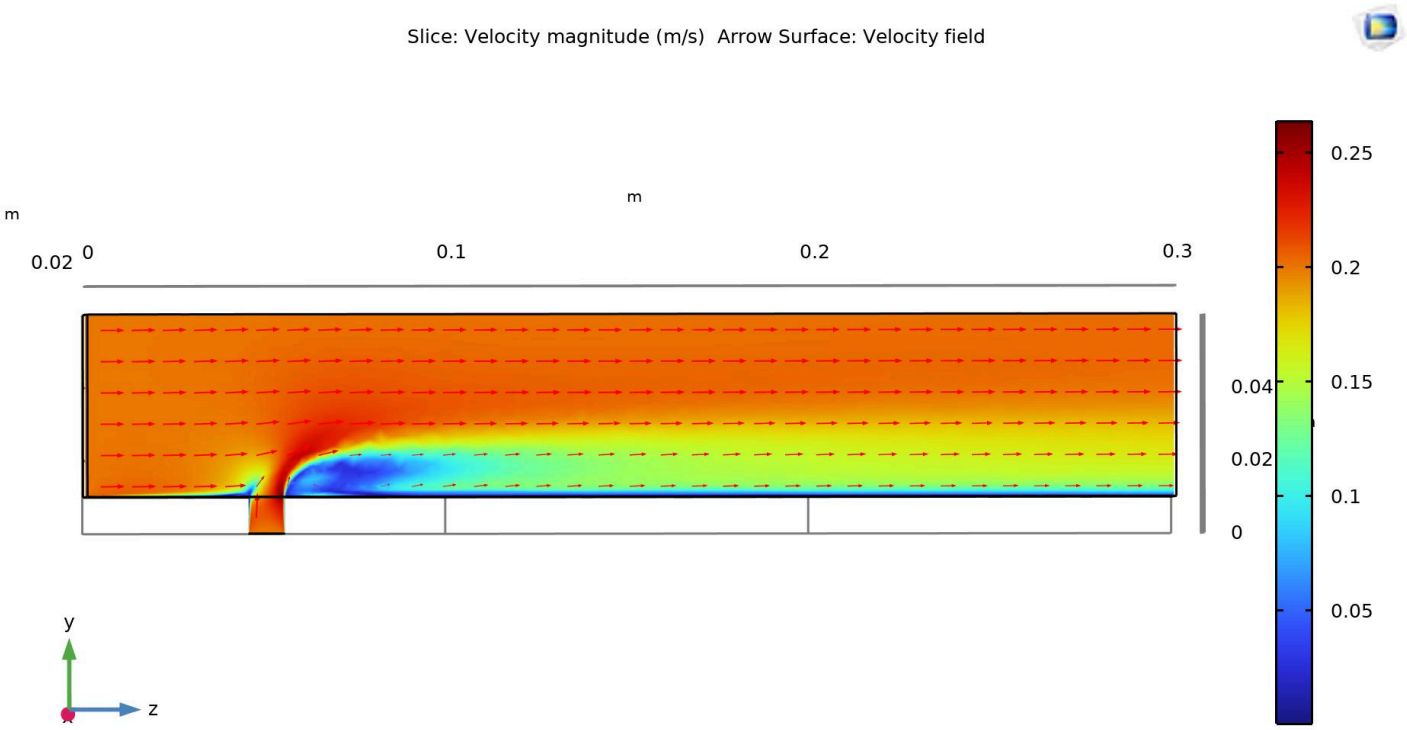
hole where fluid enters perpendicularly to the flow stream in the volume. The most notable features of this flow are the large separated region downstream of the breakout cooling hole, the separated region just before the hole, and the faster flow speeds above the separated region.

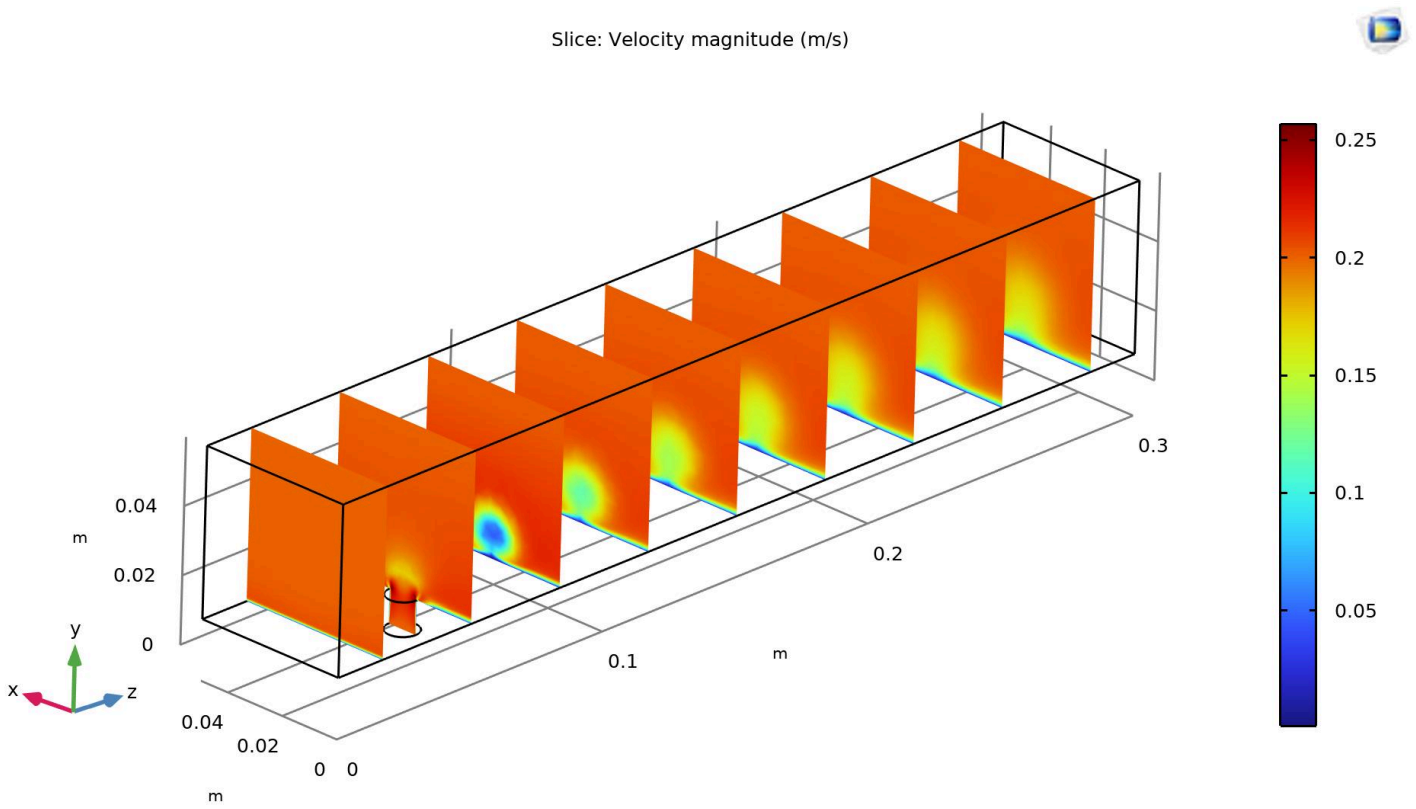
Implications for Film Cooling:

The velocity profile suggests that the cooler air does not hug the blade closely after exiting the breakout hole due to the larger separated region downstream of the breakout hole. Rather, it rises upwards into the main flow, redirecting the main stream away from the blade wall and preventing a cooling film from forming, instead creating the separated region in this area. Since the cool air lifts from the wall, much of the surface is exposed to hot main flow. The cool air spreads out, but creates a tube-shaped separated region downstream of the hole, vulnerable to hot air. The design could be improved by using a more fan-like hole opening that promotes horizontal flow of the cool air stream rather than entering the mainstream completely vertically.

Deliverable 4: k-ω Turbulence Model Comparison

Velocity Plots:





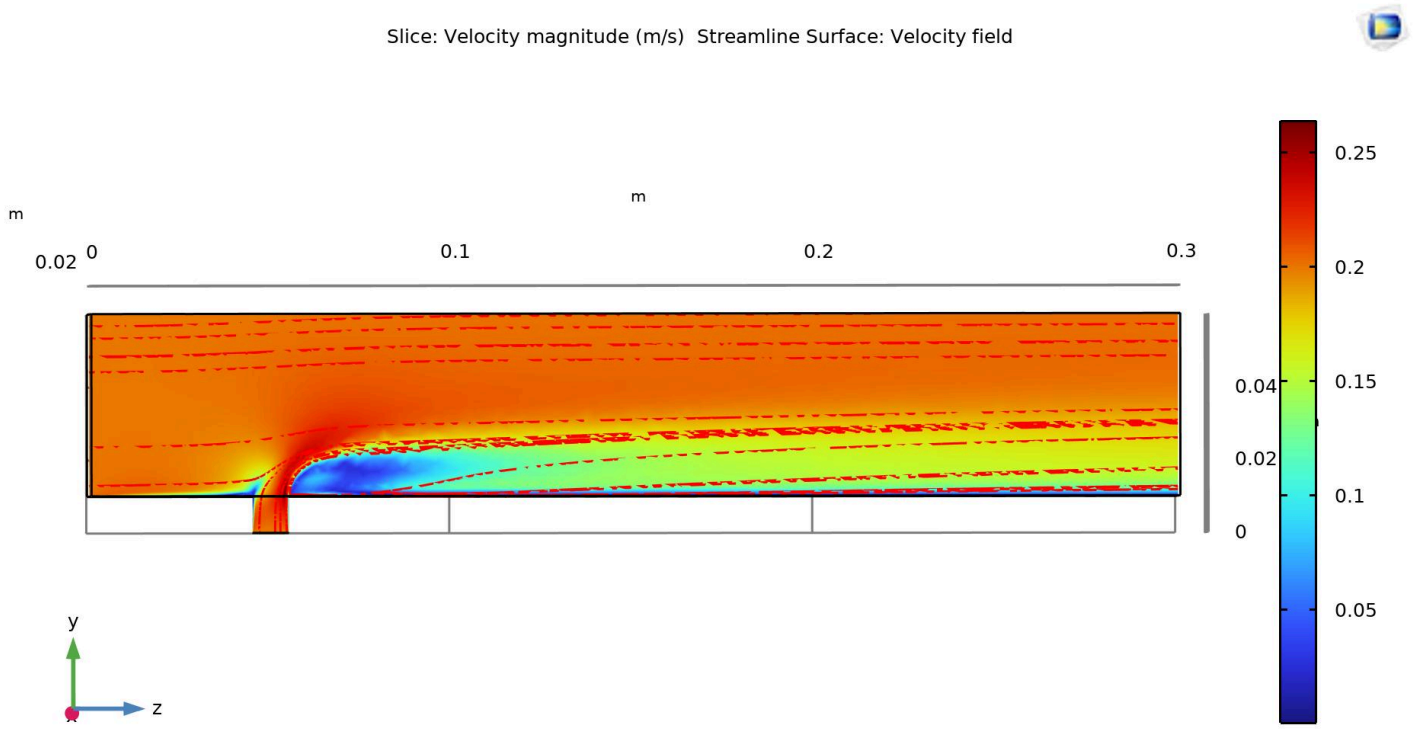


Table Comparing the k-ε and k-ω turbulence models:

	k-ε Model	k-ω Model
Maximum Velocity (magnitude)	0.28617 m/s	0.270741 m/s
Maximum Pressure on Bottom Wall	16.948 Pa	13.713 Pa

Minimum z-Direction Viscous Force on Bottom Wall	-0.44958 N/m^2	-1.3189 N/m^2
Maximum z-Direction Viscous Force on Bottom Wall	0.24682 N/m^2	0.62973 N/m^2
Average z-Direction Viscous Force on Bottom Wall	-0.15374 N/m^2	-0.17227 N/m^2

Description of Similarities and Differences of the Models:

Both models predict a similar flow pattern with separated regions and areas of high flow speed in the same locations. In both models, there is a larger separated region downstream of the breakout hole and another smaller separated region upstream of the breakout hole where bulk flow is redirected. Additionally, both have high speed regions above the separated region and relatively undisturbed bulk flow far above the breakout hole and flow regains uniformity far downstream from the hole. Major differences occur near the wall of the blade. The k-omega model provides a more accurate representation of the near-wall behavior, since it designates zero velocity at the wall due to no slip while velocity near the wall for the k-epsilon model has non-zero value. Additionally, the near-wall velocity for the k-omega model seems to be sharper higher resolution than that of the k-epsilon model. I believe this explains the disconnected streamlines in the third plot of the k-omega model since the model is better accounting for localized changes in flow. The k-epsilon model's separated region reattaches earlier (velocity vectors begin to point to the right closer to the breakout hole, reverse flow eliminated sooner) while the k-omega model's low velocity region is more pronounced and prolonged, indicating this model more effectively evaluates velocity at the near-wall regions.

Discussion of Error on Convergence Plot:

An error of 10^{-4} on the convergence plot does not mean simulation results are accurate within 0.01% overall but rather refers to the numerical difference between the left and right sides of the equations discretized over the mesh for the turbulence model. With each calculation, for example, COMSOL updates pressure and velocity and re-enters them in the Navier-Stokes equation and measures how close the left and right hand sides match. This indicates how well the equations are being satisfied and once this value is sufficiently low after a number of iterations, the solver stops, indicating that the iterative solver has converged to a stable solution. This can be seen because while both the k-epsilon and k-omega models converged to the same 10^{-4} error value, their flow fields look considerably different, particularly near the bottom wall. This shows how the error is not with respect to the true real-world velocity profile of this system. Differing plots for these two models result from different assumptions and limitations of each model.,

Deliverable 5: Turbulence Model of Choice (Algebraic yPlus)

Model Selection and Description:

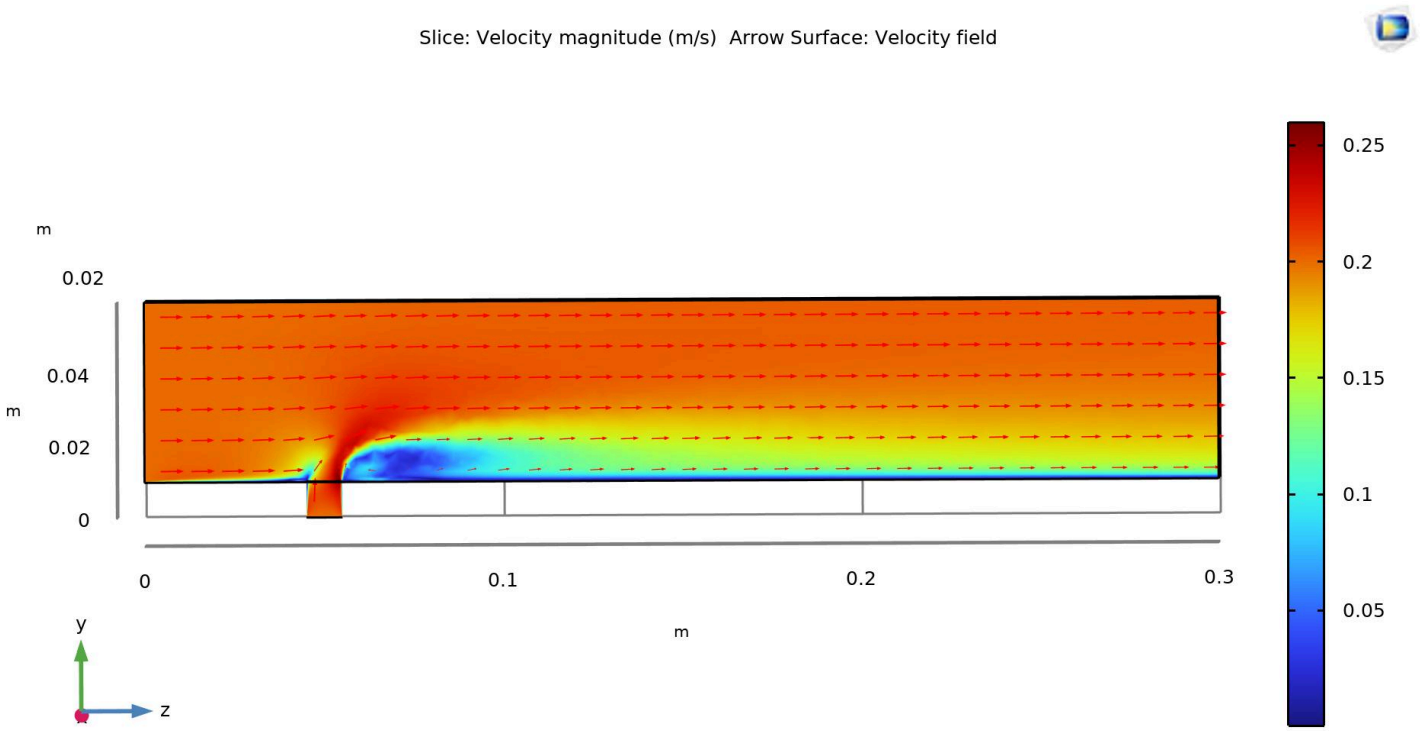
I chose to investigate the Algebraic yPlus Turbulence Model which is efficient and good for simulating turbulent flows with high Reynolds numbers. It solves the Navier-Stokes continuity and momentum equations and uses the local wall distance (dimensionless) to calculate turbulent viscosity in order to account for the turbulent flow.

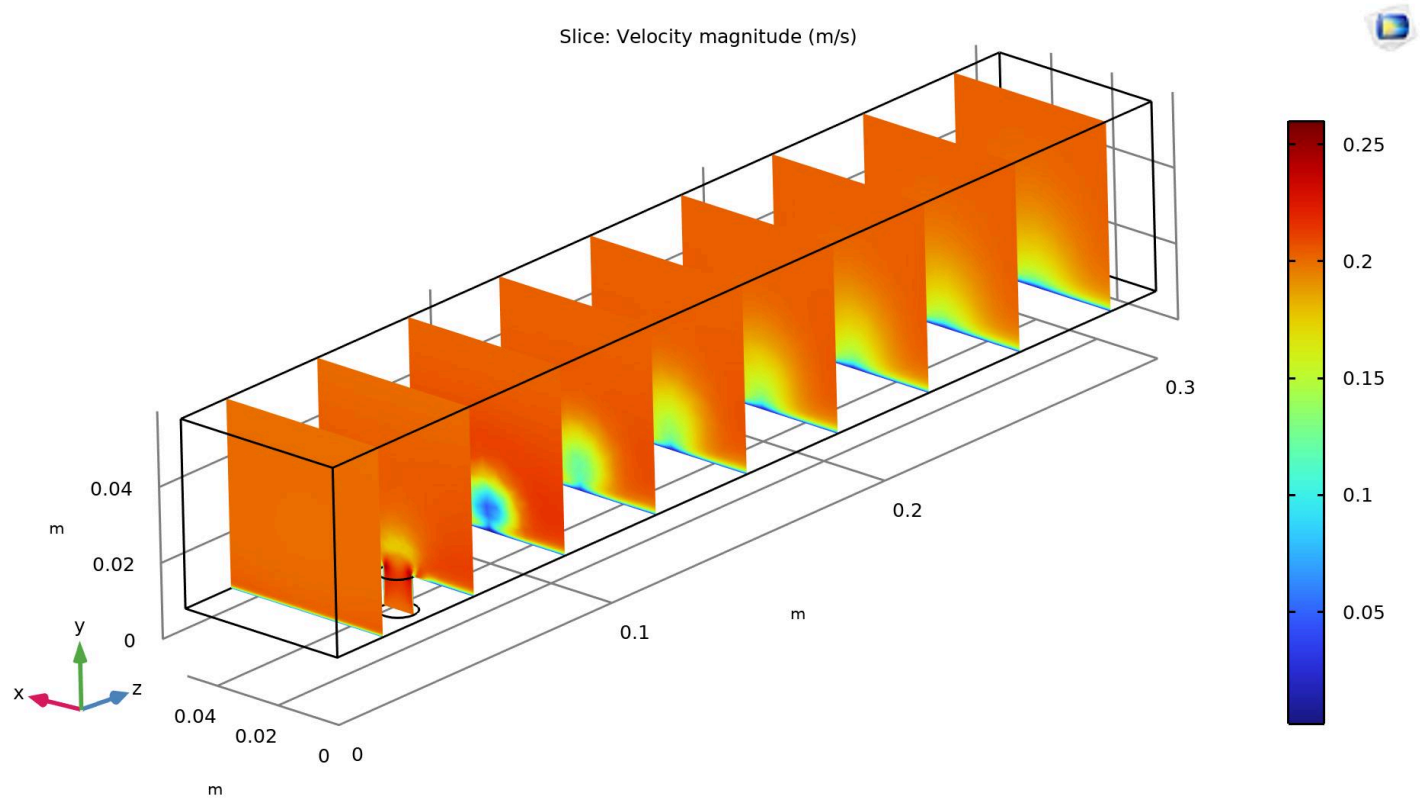
Source: https://doc.comsol.com/5.5/doc/com.comsol.help.cfd/cfd_ug_fluidflow_single.06.086.html#:~:text=The%20algebraic%20yPlus%20turbulence%20model%20is%20an%20algebraic%20turbulence%20model,(4%2D79)

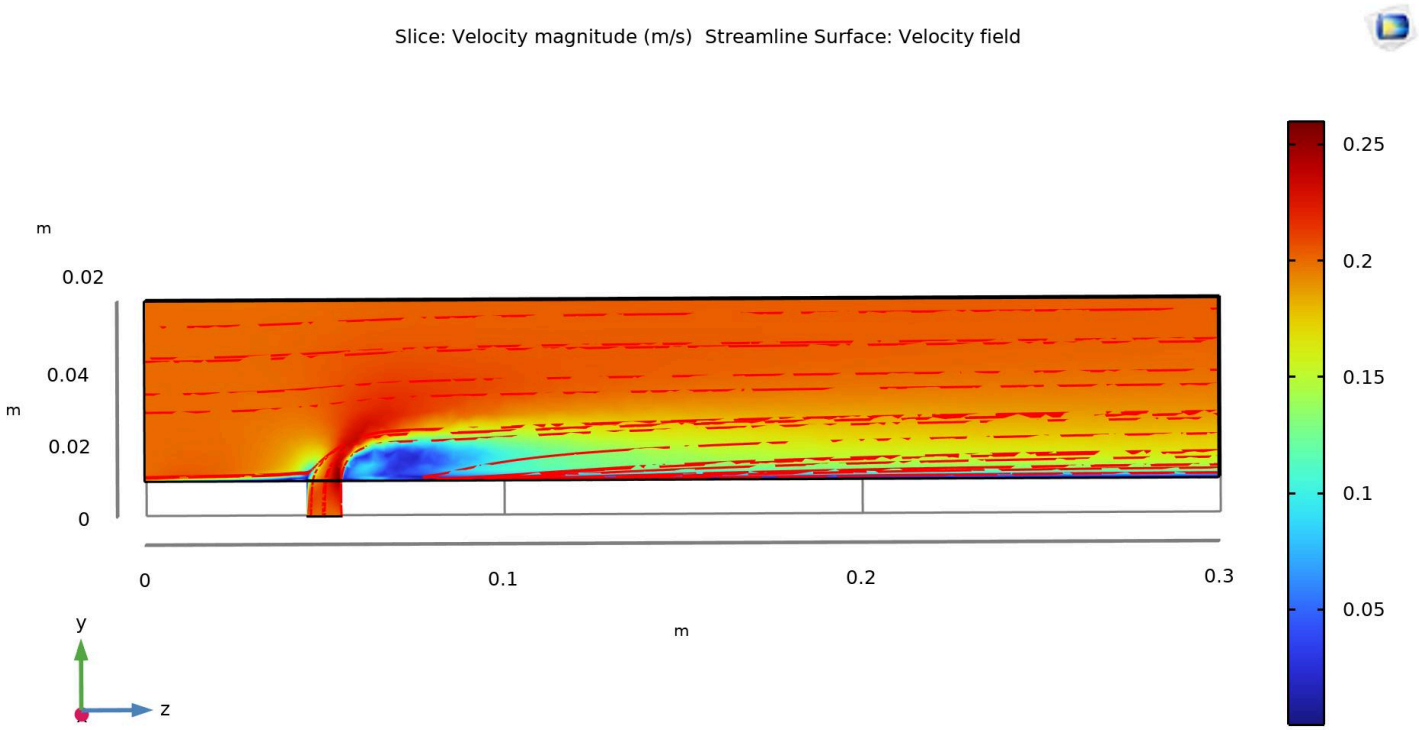
Comparison of Process with Other Turbulence Models:

Computation time was significantly reduced for the Algebraic yPlus compared to the k-epsilon and k-omega turbulence models. It took about half the time to arrive at a solution: while the k-epsilon and k-omega models took around 3 minutes and 31 seconds and 3 minutes and 9 seconds to converge respectively, the Algebraic yPlus model took only 1 minute and 25 seconds.

Velocity Plots of Choice:







Summary of Results Comparison:

The results for the Algebraic yPlus model, despite the significant reduction in computation time, are very similar to that of the k-omega model. The no slip condition and separated region near the wall is well-accounted for and the velocity profile is nearly identical to the k-omega model. The only difference is that the separated region reattaches slightly sooner for the yPlus model (as can be seen in the second plot where the first slice after the breakout hole, the velocity is higher for the yPlus model compared to the k-omega). Overall, I think the k-omega model is slightly better at handling the near-wall computations, however the two models agree much more than I expected given the significantly shorter computation time. The similarity between these models is really easily seen in the table below:

Updated Table:

	k-ε Model	k-ω Model	Algebraic yPlus Model
Maximum Velocity (magnitude)	0.28617 m/s	0.270741 m/s	0.26686 m/s
Maximum Pressure on Bottom Wall	16.948 Pa	13.713 Pa	13.973 Pa
Minimum z-Direction Viscous Force on Bottom Wall	-0.44958 N/m^2	-1.3189 N/m^2	-1.1831 N/m^2
Maximum z-Direction Viscous Force on Bottom Wall	0.24682 N/m^2	0.62973 N/m^2	0.52808 N/m^2
Average z-Direction Viscous Force on Bottom Wall	-0.15374 N/m^2	-0.17227 N/m^2	-0.18053 N/m^2

Deliverable 6: Reflection

My process for this lab involved running the simulations and reflecting on the results and completing the deliverables as I went. After doing all the supplementary reading, to begin, I had some trouble understanding the geometry and ended up having to revise the reflection on the geometry, but I understood the geometry a lot better after setting up the boundary conditions and running the simulations. I had no issues in terms of actually running the simulations. I did the entire lab in the EPDC which probably contributed to me not running into any COMSOL issues. When it came to reflecting on and comparing the results of the models, I found it challenging to compare the results from the different models and find physical explanations for certain results. I found it really helped, especially when analyzing the separated regions, to look at the vector field in understanding the velocity since this gives direction and magnitude data. I wish I had played a little more with the amount of vectors in the vector field; I may have been able to get a more detailed understanding of the flow this way. I also found that the plots looked pretty similar between the different models and it was hard to contrast them with one another, but looking at the vector field helped me find some subtle distinctions. In short, I found the modeling itself to be pretty easy while the analysis of the results was more challenging.

One lingering question I have is regarding COMSOL's mode of measuring the convergence of the solution. I did a lot of reading about this to try to understand, and learned that the error measures the discrepancy between the left- and right-hand sides of the governing equation, where the data from each iteration is re-plugged into the right-hand side and 'error' with the left hand side is measured. However, I am confused as to how we arrive at the left-hand side solution for comparison. Is this the 'true value' and if so, how is it computed? A lot of the sources I read from made it clear that it was not comparing data from successive iterations to calculate convergence like I previously thought, so I was confused as to what exactly the iterative data is being compared with to arrive at a metric for convergence of the solution.

I also have some lingering questions about how these models are developed. The supplementary reading seemed to imply that significant guesswork was involved, which is somewhat unsatisfying to me. Especially since the results from the three different models I used are very similar, I am curious what parts of the models involved guesswork since they seem pretty accurate relative to one another. I also wonder how the model predictions compared to real-world turbulent flow and how a flow situation like this could be measured empirically.

(I would also be very interested to learn more about the math for the RANS equations and how the Reynolds stresses arise as a result of time averaging velocities and more about how the Navier-Stokes equations are applied over the mesh in this type of modelling.)